

29. Data privacy is the proper handling, processing and protection of individuals' personal information to ensure it is not misused, accessed without authorization, or exposed to risks.

* GDPR is a comprehensive data protection law enacted by the European Union that came into effect on May 25, 2018.

30. Data ethics is the moral principles that guide the responsible collection, analysis, and use of data and AI technologies.

→ Using AI for discriminatory hiring practices: When AI algorithms are trained on biased data, they can perpetuate discrimination against certain groups, such as minorities or women, in hiring decisions. This is problematic because it reinforces societal inequalities.

→ Invasive data collection without consent: Collecting personal data without explicit user consent, such as tracking online behaviour or biometric data, breaches privacy rights and erodes trust. This practice is problematic because it disregards individual autonomy and can lead to misuse or unauthorized sharing of sensitive information.

→ Bias in AI decision-making system: AI systems used in criminal justice or credit scoring that produce biased outcomes can unfairly disadvantage certain populations. This is problematic because it can cause social harm, violate fairness principles and lead to unjust treatment of individuals.

31 Data governance is a structured framework that enables organizations to manage their data assets effectively throughout their lifecycle.

→ It involves establishing policies, roles, processes and standards to ensure data is accurate, consistent, secure, available, and used responsibly.

32. Bias in AI refers to systematic and repeatable tendencies within an AI system to produce unfair or discriminatory outcomes often favoring certain groups over others based on characteristics like race, gender or ethnicity.

* The consequences of bias in AI can be significant, including legal liabilities, damage to a company's reputation, and the exclusion of certain groups from opportunities or services.

33. A data analyst focuses on interpreting data to extract meaningful insights, often using tools like SQL, Excel and visualization software such as Tableau or Power BI.

A data scientist builds predictive models and advanced algorithms to analyze complex data sets, requiring skills in statistical analysis, machine learning, and programming languages like Python or R.

A data engineer designs, constructs, and maintains the infrastructure—such as data pipelines and databases that allow data collection, storage, and processing at scale.

An AI engineer (or machine learning engineer) specializes in deploying AI models into production environments, creating ~~build~~ scalable AI systems, and working with frameworks like TensorFlow or PyTorch.

REFERENCES.

1. YouTube
2. AI mode
3. Google, The Internet
4. IT SPECIALIST
5. Pub med
6. Books
7. personal Communication
8. Newspapers.

34. Business Intelligence Refers to the Strategies, technologies and processes that organization use to Collect, manage, analyze and Interpret data to Support better decision-making.

* Looker provides a unified view of a company's data Facilitating analysis across different datasets and Supporting Strategic decisions through Visualizations and Reports.

* Qlik offers a Comprehensive platform for collecting, analyzing and presenting business data, helping organisations Interpret large data sets and Identify actionable insights.

35. Artificial Intelligence and machine learning.

Because it is Impossible to go about a day without using or hearing about AI, with AI, a person can always multi-task and it is opening up a world of opportunities.

36. APIs, Tools & Developer Concepts.

→ AI mode

- Google

- YouTube.

37. Data and AI are Shifting roles from manual execution to oversight and Strategic Interpretation, forcing a move towards Continuous reskilling to remain relevant.